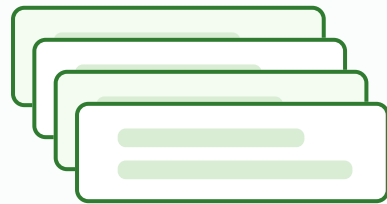


1 Experiment definition



partitioned dataset

client shards preserve data heterogeneity for repeatable comparisons

run config

task, model, method, seed, local epochs

deploy config

typed clients, resources, netem, placement

separates experiment from execution

2 Submission and admission

submit service

archive

project source + resolved configs

queue record

status, owner, priority

admission policy

capacity + oversubscribe

queue gates experiments

3 Nomad and Flower runtime

training loop: model dispatch, client update, aggregation

client i : $\theta_t \rightarrow \text{Train}_i(\theta_t) \rightarrow \Delta_i$ | server: $\theta_{t+1} = \text{Agg}(\theta_t, \{\Delta_i\})$

SuperLink + ServerApp

global θ_t , aggregation rule, arrival state

Δ_i return

θ_t dispatch

netem profile: delay / loss / jitter

rpi4

slower client tier

Δ_i

rpi5

faster client tier

Δ_i

mixed

typed slice / logical clients

Δ_i

4 Evidence surfaces

metrics

accuracy, target time, client trips

W&B metrics

logs + artifacts

analysis-ready outputs

tables, curves, diagnostics